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Five Frictions

Key Labor Market Barriers to Unlocking Job Growth in the Green Transition

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Abstract

The labor market is undergoing major changes driven by technological, economic, and demographic factors. Climate change and climate action are contributing to these shifts, driving growth in some sectors while causing decline in others. In the context of the green transition, the overall impact on employment is expected to be neutral or net positive. However, labor market frictions can hinder workers from transitioning out of declining sectors or into growing ones, posing significant development challenges. These bottlenecks can slow down the pace of the green transition and lead to adverse outcomes for workers who are unable to find suitable alternative employment, resulting in negative impacts at both the micro and macroeconomic levels.

This paper proposes a framework that classifies labor market frictions along five dimensions: what workers do, where workers are, when workers are available, who workers are, and why people work. Frictions arise when there is a misalignment between labor supply and labor demand in any of these dimensions. Within the framework, these misalignments are categorized as skill-, spatial-, temporal-, norm-, or preference-related mismatches, respectively. Drawing on insights from World Bank analyses, the paper further identifies potential solutions to address each friction, providing guidance for policymakers to facilitate smoother workforce transitions and maximize macroeconomic benefits from the green transition. Although developed in the context of the green transition, the framework can be generalized to other economic shocks and transformations.

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Produced by the Research Support Team

Five Frictions: Key Labor Market Barriers to Unlocking Job Growth in the Green Transition*

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1. Introduction

The green transition, characterized by a shift towards environmentally sustainable practices and technologies, is reshaping the global economic landscape. This transition is driving growth in certain sectors, such as renewable energy and green technologies, while leading to declines in others, such as fossil fuels. These sectoral shifts have profound implications for labor demand, influencing people's jobs and livelihoods in significant ways.

Research indicates that the impact of green transition policies on employment tends to be neutral or net positive (Godinho 2022; Lehr and Politt 2024; Hanna et al. 2024). For example, the International Labour Organization estimates that the green transition in the energy sector could create 24 million new jobs globally by 2030, offsetting the loss of 6 million jobs (ILO 2018). However, these analyses often operate under the assumption that labor markets are highly efficient and that workers can transition from declining sectors to those experiencing growth. This assumption overlooks the complexities and frictions inherent in real-world labor markets.

In practice, labor market frictions can pose significant challenges for workers attempting to transition out of declining sectors or into growing ones.[1] These frictions include mismatches in skills, location, timing, norms, and preferences—either individually or, often, in combination. As a result, workers may find it difficult to transition from jobs with decreasing demand to jobs with increasing demand, leading to labor market bottlenecks. These bottlenecks can slow down the pace of the green transition, as sectors experiencing growth may struggle to find the skilled labor they need to expand. In addition, these labor market dynamics can lead to higher unemployment and adverse outcomes for workers who are unable to find alternative employment opportunities. This can exacerbate political economy challenges, as affected workers and communities may resist interventions aimed at addressing environmental and climate issues due to fears of job loss and economic instability (Hallegatte et al. 2023).

This paper proposes a framework for identifying labor market frictions that are critical to unlocking job growth in the green transition and suggests possible solutions to address each barrier. Recognizing that there is no one-size-fits-all solution, it provides indicative measures of labor market adjustments and points to literature that explores these options in greater depth. By examining the specific challenges faced by workers and proposing targeted interventions, this paper seeks to facilitate smoother and more inclusive transitions for the labor force, thereby supporting the overall goals of the green transition. The paper draws on several World Bank analyses, particularly from Country Climate and Development Reports (CCDRs),[2] as well as key literature.

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[1] While the arguments in this paper focus on sectoral shifts, the green transition is likely to also entail within-sector firm reallocation (Arias et al. 2025). For example, more technologically advanced firms tend to be more energy-efficient and therefore better positioned to adapt to the green transition (Comin et al. 2022). The proposed framework, however, applies at both margins.

[2] CCDRs examine country transition pathways, derived from a combination of sectoral and macroeconomic modeling, and provide policy recommendations to support resilient low-emission development. These recommendations include strategies to create jobs as well as targeted supply-side interventions that can help ensure the workforce aligns with the evolving needs of the labor market. Read more at: https://www.worldbank.org/en/publication/country-climate-development-reports.

While it does not provide an exhaustive review of all existing literature, it focuses on offering actionable insights to a general policy audience.

The proposed framework is presented in the context of the green transition, but it generalizes to other economic shocks and transitions, such as the impact of technological change (e.g., automation and artificial intelligence) or demographic change (e.g., workforce aging). This is because the challenge is not merely to transition from a subset of jobs that are "brown" to a subset of jobs that are "green", but to adapt to broader changes in labor demand and supply. As discussed in our companion paper (Mealy et al., forthcoming), defining and measuring green jobs is conceptually and empirically challenging, and focusing narrowly on green jobs can obscure the wider and more dynamic employment reallocation underway. Instead, Mealy et al. propose identifying "jobs in demand" and "jobs at risk" under various country-specific green transition scenarios. A key appeal of this framing is that it remains useful even amid multiple, concurrent transformations. For example, technological advancements can create new job opportunities while rendering certain skills obsolete, and demographic shifts, such as an aging population, can alter the composition and availability of the workforce. Adopting a jobs-in-demand/jobs-at-risk lens thus provides a more comprehensive and flexible approach to diagnosing barriers to worker mobility.

Importantly, it is often neither feasible nor desirable to expect workers in occupations at risk to transition directly into occupations in demand. For example, a coal miner may not readily become a solar panel installer due to differences in skills, experience, or geographic location. Instead, the green transition and other concurrent shifts are more likely to take place through a series of incremental moves, resulting in a broader, multi-step reshuffling of workers. Network analysis offers a valuable approach for examining these flows, and this paper presents several examples to illustrate how network analysis can be used to identify feasible transition pathways. Integrating network flows with macroeconomic models could further enhance our understanding of these dynamics. Mealy et al. (forthcoming) review emerging approaches to enhancing labor market dynamics in macroeconomic models, including the use of network analysis and agent-based modeling to reflect the full spectrum of worker reallocation.

The remainder of the paper is organized as follows. Section 2 presents the framework for identifying key labor market frictions. Section 3 complements this framework with examples of analytical work related to each friction. Section 4 explores potential policy options to mitigate these frictions. Finally, section 5 concludes with key takeaways and directions for future research. Together, these sections provide practical insights for addressing labor market challenges and promoting job growth in the context of the green transition.

2. A framework for identifying labor market frictions in the green transition

A typology of labor market frictions can serve as a useful first step in developing effective policies and programs to support workers through the green transition. This paper proposes a framework that classifies challenges in worker mobility along five dimensions: *what* workers do, *where* workers are, *when* workers are available, *who* workers are, and *why* people work (table 1). For each dimension, the

framework poses broad questions with the aim to understand the needs of employers (labor demand) compared with the availability of workers (labor supply). Mismatches occur when there is a difference between labor demand and labor supply. These include:

- **Skill mismatches:** When there are differences between the skills that employers need and the skills that workers offer.
- **Spatial mismatches**: When there are differences between where jobs are available and where workers are located.
- **Temporal mismatches**: When there are differences between when workers are needed and when workers are available.
- **Norm mismatches**: When there are disparities between the biases of employers and the characteristics of workers.
- **Preference mismatches**: When there are differences between the preferences of workers and the job attributes offered by employers (including, but not limited to, compensation).

This framework builds on existing knowledge and presents known labor market frictions in a simple and structured manner. Despite the inherent complexities of labor market dynamics, the framework is designed to assist policymakers in pinpointing specific areas where interventions are needed. By systematically categorizing different types of mismatches, it aids practitioners in identifying key issues, thereby enhancing their ability to develop targeted strategies to address the multifaceted nature of these challenges. Crucially, these frictions originate from a variety of factors, including transaction costs (such as relocation expenses) and market failures (such as information asymmetries). The effectiveness and efficiency of policies and interventions aimed at reducing the frictions depend on a rigorous identification of their underlying causes, ensuring that solutions are appropriately tailored to address the root issues.

While the framework presents the five mismatches independently, in practice, they often co-occur. For example, skill mismatches can intersect with norm-based frictions. Women may face barriers to accessing education and training programs in certain fields, leading to a gender disparity in skills that are in high demand. This can result in women being underrepresented in sectors such as engineering, where there is a significant demand for skilled workers. In the framework, each friction is presented separately for clarity and to acknowledge that each friction can be a standalone issue, but several analytical examples in the next section examine the interactions of multiple frictions, highlighting the often complex and interconnected nature of these challenges.

Additionally, the boundaries between the five frictions are not always clear-cut. For example, spatial frictions can overlap with frictions related to workers' preferences—such as their tolerance for commute time. Likewise, distinguishing between mismatches driven by social norms and those driven by individual preferences can be challenging. The framework separates *preferences*, which reflect individual desires or priorities, from *norms*, which are socially constructed expectations or behaviors. Compensation offers a good example: it is a core job attribute to which workers assign varying degrees of importance and preference mismatches can, for example, discourage labor transitioning if compensation is lower in emerging sectors relative to declining ones. However, compensation is also susceptible to norm-based mismatches, such as wage disparities by gender or race.

Table 1: A framework for identifying labor market frictions in the green transition

Friction	Labor demand	Labor supply	Potential mismatch	Examples of analytical work
WHAT	• *What* skills are needed to deliver the green transition? • *What* skills are likely to be displaced?	• *What* are the skills available in the workforce?	**SKILL** mismatches	In Namibia, only 6 out of 63 occupations related to the green hydrogen industry are sought by unemployed workers, indicating a significant skill gap.
WHERE	• *Where* are jobs created and what is the spatial distribution of the skill requirements? • *Where* are jobs lost and what is the spatial distribution of displaced skills?	• *Where* are workers and skills located?	**SPATIAL** mismatches	In Brazil, 88% of all formal job switches occur within the same state, making the geographic separation between jobs-in-demand and jobs-at-risk a significant barrier to workforce mobility.
WHEN	• *When* and for how long are workers and skills needed? • *When* and how fast are workers and skills being displaced?	• *When* (or how quickly) can workers and skills be mobilized?	**TEMPORAL** mismatches	In Tajikistan, about 65,000 workers will be needed for rooftop solar installations, but this surge is expected to be temporary, creating a temporal challenge as labor demand quickly tapers off and there is a risk of workforce oversupply.
WHO	• *Who* is affected by employer biases?	• *Who* make up the workforce (what are the characteristics of workers)? • *Who* is impacted by self-bias?	**NORM** mismatches	Between 2020 and 2023, only 3% of workers entering male-dominated occupations in Georgia (defined as those with at least 85% men) are women, and 7% entering female-dominated roles are men.
WHY	• What job attributes do employers offer?	• *Why* do people work (how do workers prioritize job attributes)?	**PREFERENCE** mismatches	In Malaysia, workers transitioning from at-risk oil and gas occupations to in-demand roles with similar skill requirements face wage reductions ranging from 20% to 70%.

3. Five labor market frictions

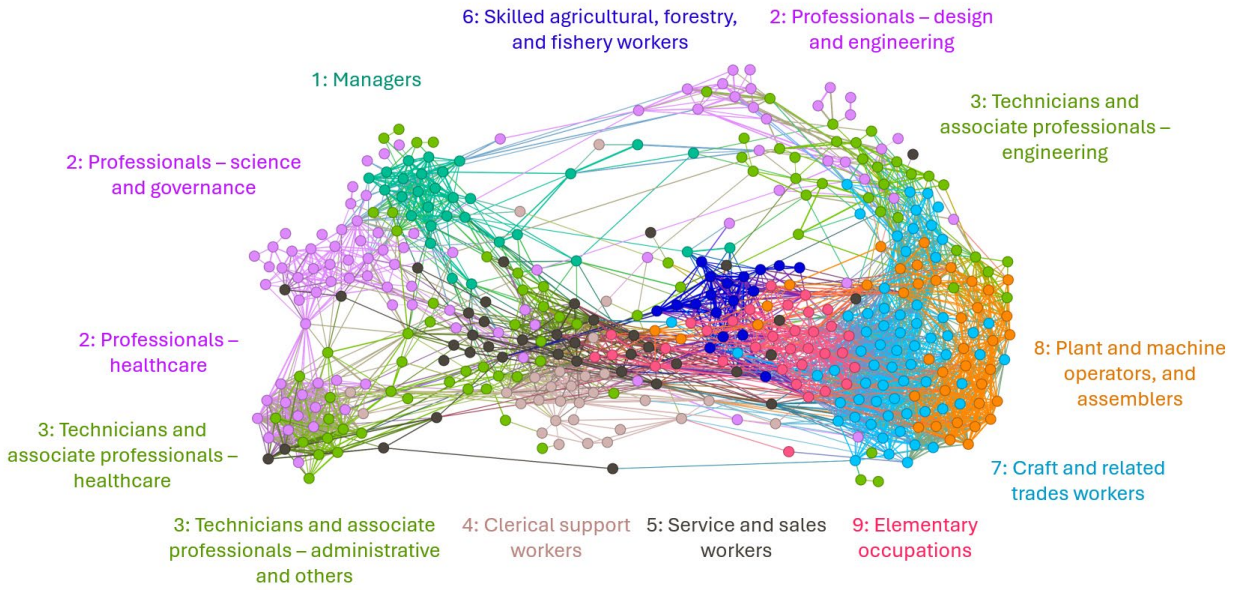
This section provides examples of analytical work that explores the labor market barriers introduced in section 2. Most of these examples draw on country-level analyses conducted as part of the World Bank Country Climate and Development Reports (CCDRs), but they also incorporate key insights from other literature. Although many examples focus on the energy sector, their main objective is to illustrate analytical approaches that can be applied to examine each of the five barriers. The methodologies differ across examples, reflecting differences in country contexts and data availability. Detailed descriptions of the methodologies are available in the original studies, and related methodological challenges are discussed in Mealy et al. (forthcoming).

3.1. Skill mismatches

Skill mismatches occur when the skills available in the workforce are different from those required by employers. Several recent studies have made significant strides towards quantifying skill mismatches using methods that rely on empirically observed mobility between occupations or industries (e.g., Berryman et al. 2025, del Rio-Chanona et al. 2021, Lankhuizen et al. 2022) or a measure of occupational relatedness based on tasks or skills (Aufiero et al. 2024, Bowen et al. 2018, Bücker et al. 2025, Frank et al. 2024, Mealy et al. 2018). Both approaches have pros and cons. Relying on past job-switching data provides a straightforward way to assess frictions, but changing norms, policies, or technologies can limit the usefulness of a purely backward-looking approach. Additionally, longitudinal labor data is often not available, especially in lower-income countries. Task- or skill-based approaches can help overcome these issues but may overlook other labor market frictions, such as those related to cultural or social norms. Moreover, different measures of task or skill similarity often yield different results (Mealy et al., forthcoming).

Whether using a backward-looking approach that utilizes historic transitions or a forward-looking approach that focuses on tasks and skills, network models (or graph theory) can offer a useful analysis and visualization tool. Figure 1 shows an example of a network based on task similarity. Nodes represent occupations that are connected if they share a large number of tasks. Occupations requiring similar tasks are positioned closer together, while those with less similarity are placed further apart. Nodes are colored according to their one-digit occupation family, with higher-skilled occupations predominantly located in the left and upper-right regions, and lower-skilled occupations mainly in the lower-mid and lower-right regions. This suggests that occupations with similar skill levels tend to involve some degree of similar tasks. For example, senior government officials and urban planners, both high-skilled professions, share tasks such as liaising with local authorities and applying civic skills. Similarly, wood treaters and chemical machine operators, both low-skilled professions, share several tasks such as disposing of hazardous waste, working safely with chemicals, and inspecting industrial equipment.

Figure 1: A network visualization showing high task similarity between occupations with similar skill levels



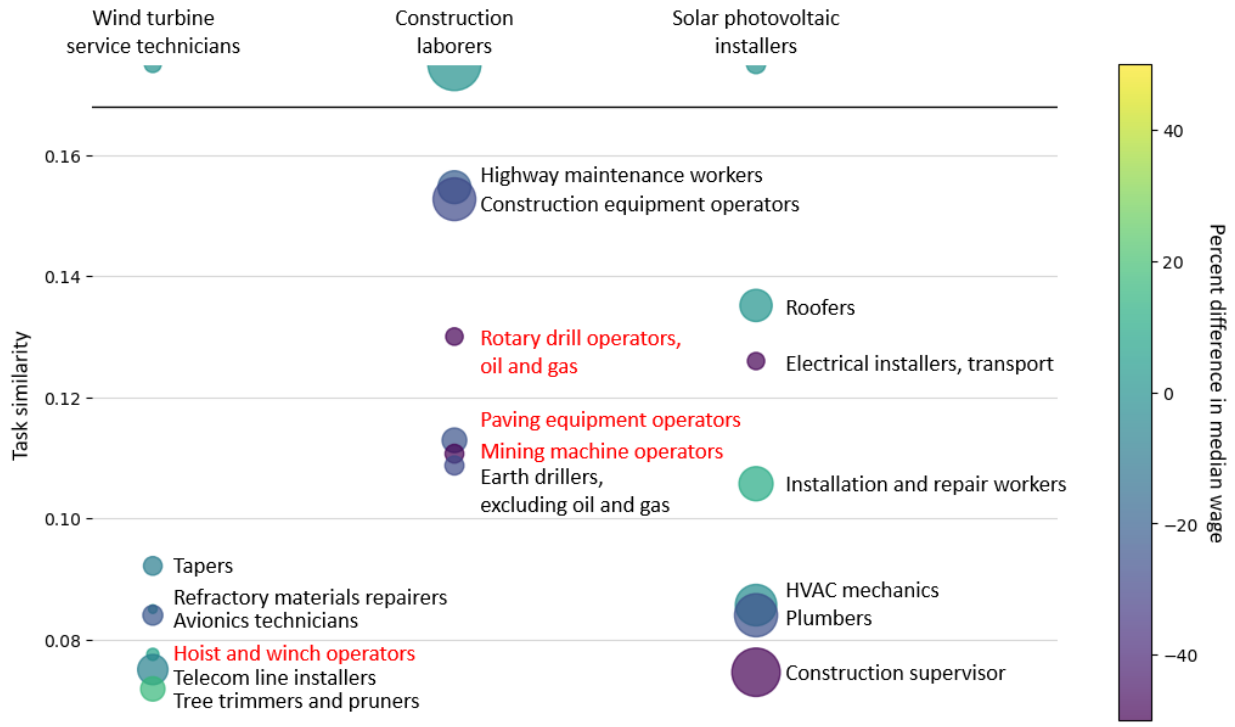
Source: Adapted from World Bank Group 2025.
Notes: Nodes represent occupations and are colored according to their one-digit occupation family, following the International Standard Classification of Occupations (ISCO-08).

Several CCDRs use measures of occupational similarity to better understand what workers in different occupations do and, importantly, the occupations they *could* do with minimal retraining. The underlying assumption is that workers are more likely to transition into occupations that require similar tasks to their current profession (Mealy et al. 2018). As such, these analyses offer valuable insights that not only highlight critical bottlenecks but also provide actionable information for designing solutions. This includes identifying specific occupations with high task similarity for roles anticipated to undergo significant changes in demand.

For example, the Uzbekistan CCDR presents a pathway in the power sector that involves a shift from fossil fuels to renewable energy sources (World Bank Group 2023a). This transition requires mobilizing a significant number of workers for the construction, installation, operation, and maintenance of new renewable energy infrastructure, as well as finding alternative roles for workers currently in the fossil fuel industry. Key occupations in demand include construction laborers, wind turbine technicians, and solar photovoltaic installers. Analysis using United States data suggests that some of these growing occupations have strong similarity with other roles (figure 2). For example, solar photovoltaic installers have a considerable task overlap with roofers and some installation and repair workers. Similarly, rotary drill operators in the oil and gas sector share 10 of the 23 tasks performed by construction laborers—including 'operate pumping systems or equipment' and 'assess characteristics of land or property'—making their transition to this role more feasible than for workers in occupations with less task overlap. Wind turbine service technicians, on the other hand, exhibit relatively weak task similarity with their closest related occupations, which include specialized roles, such as tapers, avionics technicians, hoist

and winch operators, and tree trimmers and pruners. These occupations share the specialized task of climbing equipment or structures.

Figure 2: Task similarity among growing occupations and their closest related roles

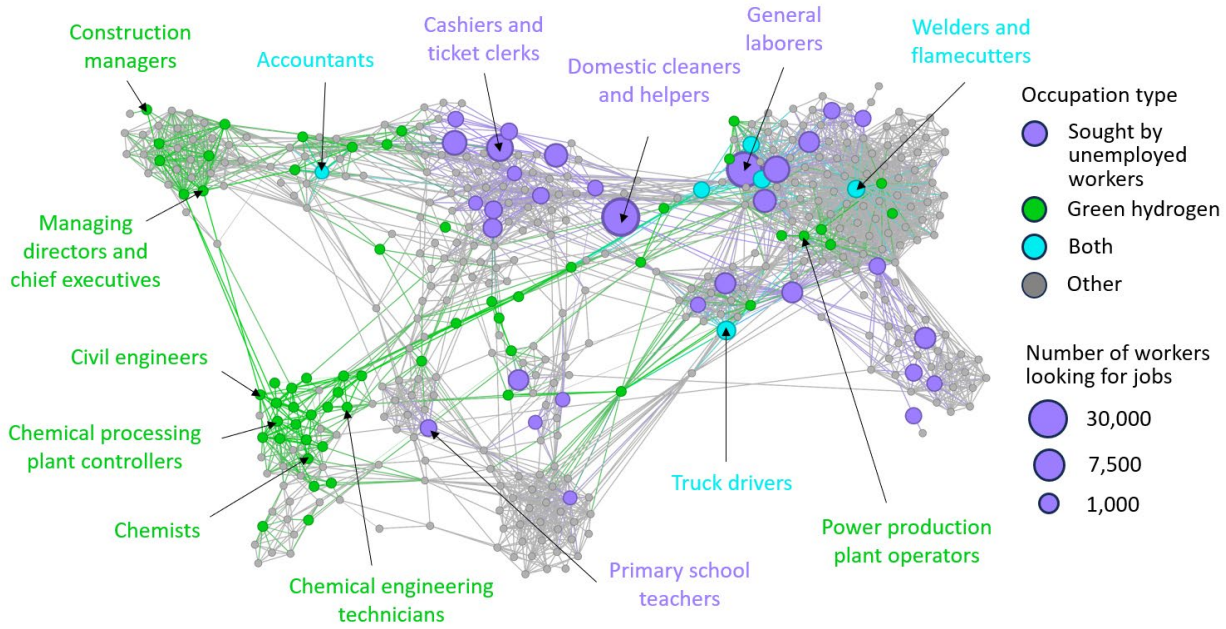


Source: Adapted from World Bank Group (2023a).
Notes: Node size indicates total employment per occupation in the United States; coloring indicates if moving to the growing occupation would be a wage increasing (yellow) or decreasing (purple) move; occupations with a red label are likely to be at risk of declining labor demand in a low-carbon scenario; task similarity is normalized to a value between 0 (no task overlap) and 1 (highest task overlap between any two unrelated occupations); HVAC = heating, ventilation, and air conditioning.

Another example from the Namibia CCDR examines the skill mismatches that could hinder or delay the development of a green hydrogen economy (World Bank Group, forthcoming-a). The analysis identifies occupations relevant for a diverse range of roles in the production, storage, and distribution of green hydrogen. Namibia, a country with structurally high unemployment, has adopted green hydrogen production as a strategic priority to create jobs and boost economic growth. However, figure 3 illustrates a significant mismatch between the occupations sought by unemployed workers (purple) and those required to support green hydrogen production (green), with only 6 out of 63 occupations overlapping (blue). Unemployed workers predominantly seek low-skilled jobs located in the upper and right areas of the network, whereas green hydrogen production demands several high-skilled jobs concentrated in the left region of the network. This disconnect indicates that the skills required for occupations in the green hydrogen sector have limited overlap with the skills possessed by most unemployed workers. Additionally, the majority of the employed population is engaged in occupations that require different skills from those demanded by the green hydrogen sector. As a result, there are relatively few workers in

the labor force whose existing skills would allow them to transition smoothly into green hydrogen occupations.

Figure 3: Occupational network showing limited overlap between the roles sought by unemployed workers and those relevant to the green hydrogen industry in Namibia

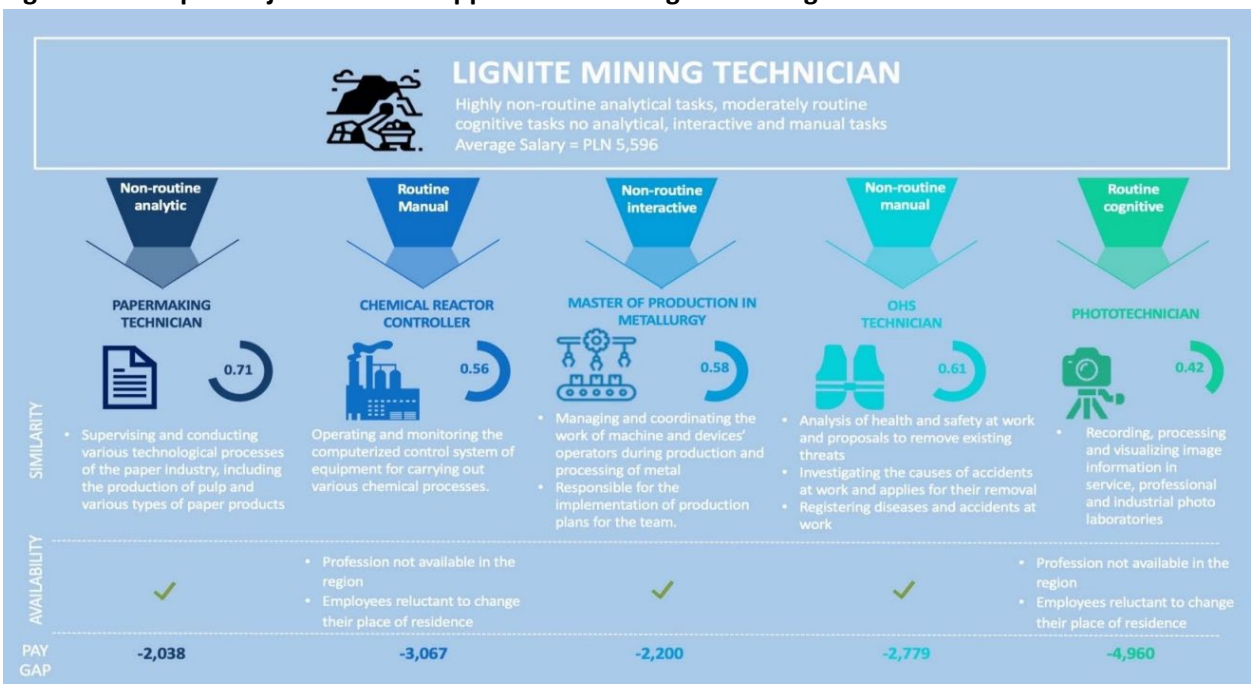


Source: World Bank Group, forthcoming-a.
Notes: Purple nodes represent occupations sought by unemployed workers; green nodes represent occupations required to support green hydrogen production; blue nodes represent occupations that fall into both categories; purple and blue nodes are sized according to the number of unemployed workers seeking jobs according to Namibia's 2018 labor force survey; green and gray nodes are of a uniform size.

The previous examples explore skill mismatches or skill gaps in the context of mobilizing workers to fill roles in high-demand occupations. However, it is also crucial to consider skill mismatches for workers in occupations with decreasing demand, as these can impact the ability of displaced individuals to transition smoothly to new roles. A notable example of this type of analysis includes a series of regional studies on the just transition of coal workers in Poland.[3] By utilizing available data on job descriptions, the analysis calculated similarity scores between occupations and used these to identify possible career switches for coal workers. For example, figure 4 illustrates five job transition opportunities for lignite mining technicians. The analysis also introduced an additional metric to assess skill similarity across occupations, evaluating the routine and non-routine task content of each occupation, based on a measure developed by Mihaylov and Tijdens (2019). Other factors considered include salary, location, and job security, which are covered in the following sections.

[3] https://www.worldbank.org/en/country/poland/publication/support-for-polish-coal-regions-in-transition

Figure 4: Examples of job transition opportunities for lignite mining technicians in Poland



Source: Christiaensen et al. (2022).
Notes: A similarity score close to zero indicates minimal task overlap between occupations, while a similarity score close to one indicates substantial task overlap between occupations.

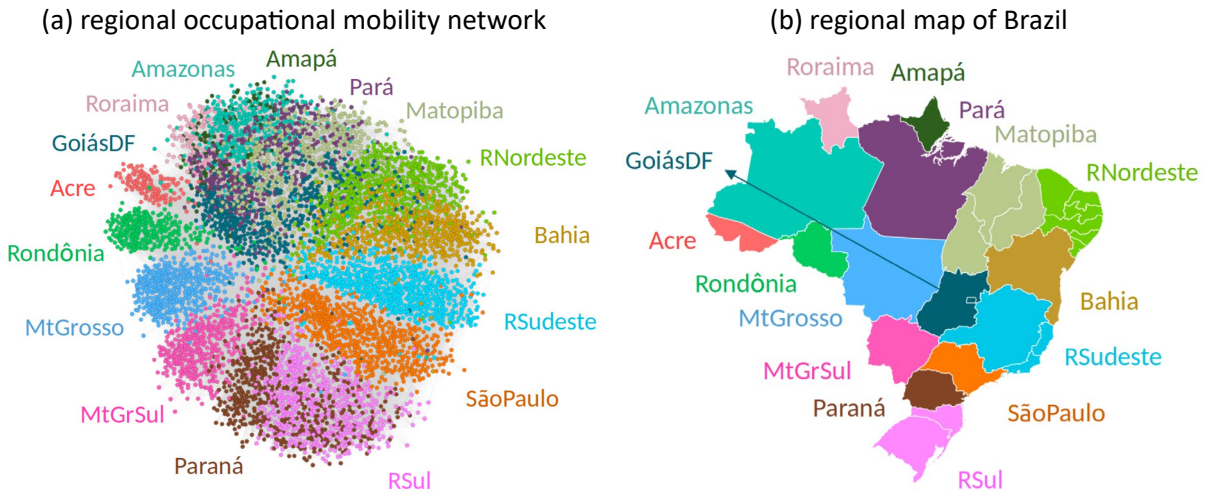
3.2. Spatial mismatches

While home-based or flexible work arrangements are becoming more common, especially for many white-collar jobs, most occupations still require workers to be physically present at a worksite. Consequently, spatial mismatches arise when there is a disparity between where labor is demanded and where it is supplied. Beyond physical distance, spatial or geographical mismatches can also be influenced by a broader range of issues, such as the availability and quality of transportation, housing, and public services. These mismatches can result in some regions experiencing underemployment while others face labor shortages, ultimately impacting national and regional economic growth and inequality. As such, while aggregate estimates of the labor impacts of the green transition are often neutral or net positive, there is a growing body of literature on the uneven distribution of employment opportunities and job losses across different locations (Saussay et al. 2024; Sharma and Banerjee 2021).

Although some workers relocate, empirical evidence indicates that most job transitions occur within the same geographic area (Bartik 2020; Berryman et al. 2025), particularly in low- and middle-income countries (Grover et al. 2022). In Brazil, about 88% of all formal job switches between 2011 and 2019 happened within the same state, as illustrated in figure 5. The left panel of the figure depicts a regional occupational mobility network, where each node represents a specific occupation within a given region. The links between nodes reflect the probability of transitioning from one occupation-region node to another. Nodes that are closer together indicate a higher probability of job transitions, while nodes that are further apart indicate a lower probability of transitions. The clustering in the network, with nodes

color-coded by region, strongly resembles the regional geographical boundaries of Brazil, as shown in the right panel. This alignment suggests that most job transitions happen locally rather than across different regions.

Figure 5: Regional occupational mobility network and map of Brazil

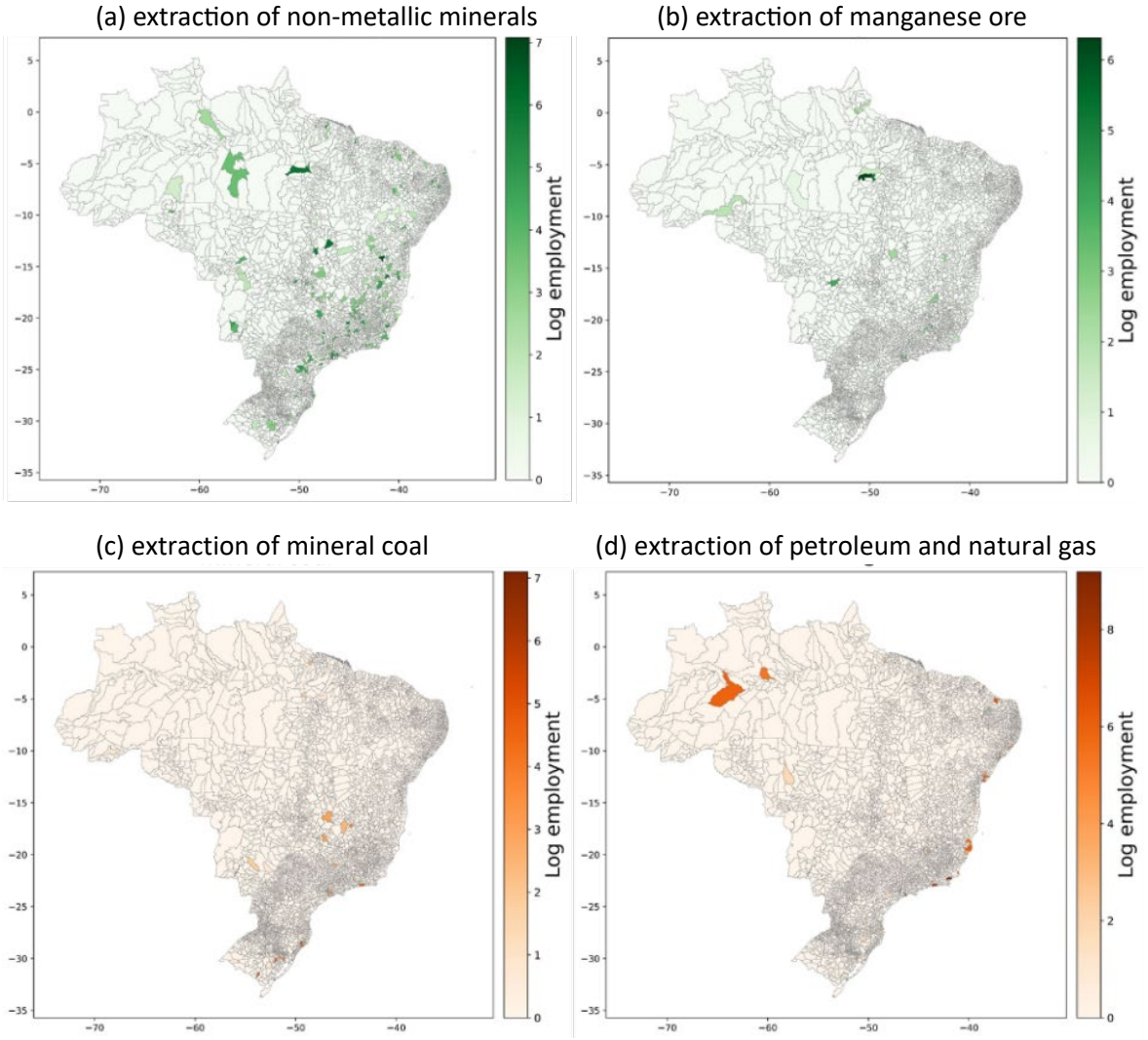


Source: Berryman et al. 2025.
Notes: In the left panel, nodes represent a specific occupation-region pair. The same occupation may appear multiple times in the network, once for each region where it is present. The nodes are color-coded by region. Links between nodes indicate the probability of transitioning from one occupation-region node to another. Nodes positioned closer together reflect a higher probability of job transitions, while those further apart reflect a lower probability. The right panel displays a map of Brazil colored by region. The analysis is based on data from the Relação Anual de Informações Sociais (RAIS) for the years 2011-2019.

Understanding spatial barriers is essential to supporting a just and inclusive green transition. For example, the Brazil CCDR found that some green jobs, such as those in green mineral extraction, require skills similar to those in some fossil fuel-related jobs, like coal mining (World Bank Group 2023b). However, while displaced coal workers may possess several of the necessary skills for green mineral extraction, the differing locations of mineral deposits can pose a significant barrier. Figure 6 illustrates the geographical distribution of workers in two green mineral (top panels) and two fossil fuel extractive industries (bottom panels) in Brazil. Although there are some overlaps, the geographical profiles of these industries differ significantly.

This challenge is not unique to Brazil; many remote mining communities around the world face similar issues. A survey of coal workers in Poland, for example, shows a high aversion among workers to relocation and even commuting (Christiaensen et al., 2022). The reduced dependence on fossil fuels can thus have profound impacts on these communities, causing multiple disruptions to jobs, household incomes, local economies closely linked to the coal supply chain, community well-being, and the capacity and fiscal health of local and regional governments (Ruppert Bulmer et al., 2021).

Figure 6: Geographical distribution of employment in selected green mineral and fossil fuel sectors in Brazil



Source: Developed by the authors.
Note: The analysis is based on data from the Relação Anual de Informações Sociais (RAIS) for the year 2018.

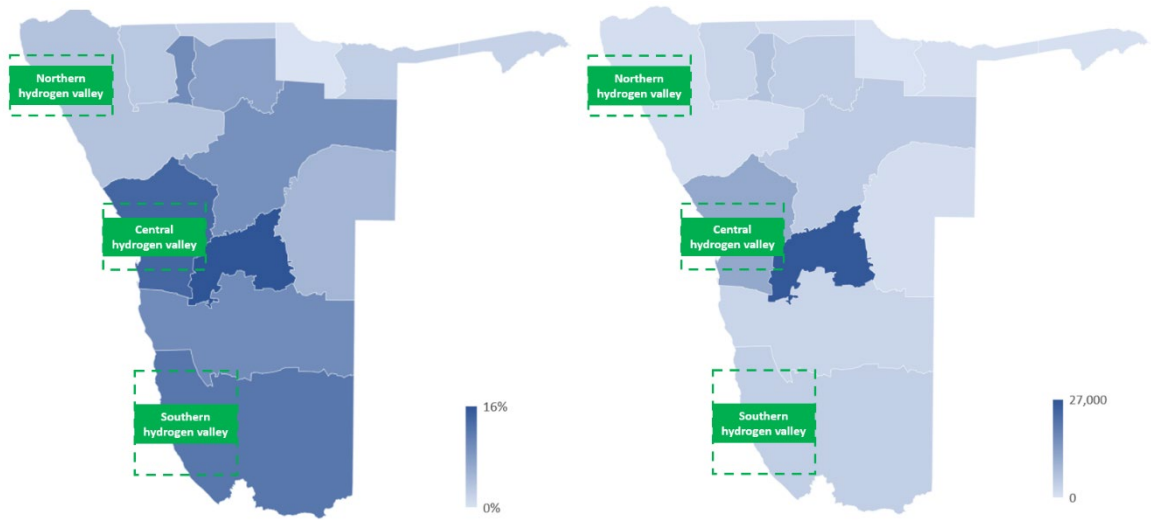
In addition to green minerals, renewable energy is often cited as a means to offset job losses in the fossil fuel industry and provide alternative employment opportunities (Hanna et al. 2024; Lehr and Ruppert Bulmer, forthcoming). However, even if local reallocation of workers could enable fossil fuel workers to find new jobs and renewable energy positions to be filled, the potential for renewable energy development to provide alternative employment opportunities in former coal mining areas is limited by geographic suitability: only 29% of coal mining sites in China and 62% in the United States are suitable for solar power (Pai et al. 2020). In both countries, less than 7% of coal mining areas have wind speeds that are suitable for wind power.

Understanding spatial barriers is also essential to ensure that workers can be mobilized into areas where labor demand is increasing, thereby avoiding labor market bottlenecks that can slow down the pace of

the green transition. The Namibia CCDR, for example, shows that spatial challenges could present a significant obstacle to the achievement of the country's green hydrogen and derivatives strategy (World Bank Group, forthcoming-a). The plan involves the development of three green hydrogen valleys. While the central hydrogen valley benefits from a relatively large concentration of workers in occupations relevant to the green hydrogen industry, particularly in the neighboring region which includes the capital Windhoek, the northern and southern valleys could face significant worker shortages (figure 7). Although the southern valley has a relatively high concentration of workers in green hydrogen-relevant occupations (left panel), the total number of workers remains low (right panel).

Figure 7: Maps of Namibia highlighting workforce scarcity in priority locations for green hydrogen production

(a) Share of total workers (b) Number of workers



Source: World Bank Group, forthcoming-a.

While spatial mismatches are often attributed to the disparity between the locations of the current workforce and job opportunities, they can also stem from gaps in education and training for new entrants into the labor market. For example, Pakistan has adopted ambitious national renewable energy targets. However, a survey of renewable energy firms and technical colleges and universities in the country found that while renewable energy education and training options are available throughout the country, they are most limited in regions with the highest wind and solar potential (World Bank 2022a). This educational gap could exacerbate spatial mismatches if individuals trained in renewable energy are reluctant to relocate from where they are trained to areas where their skills are most needed.

Spatial mismatches can occur not only at national or regional scales but also at the local level, where complex urban challenges such as inadequate infrastructure or a lack of affordable housing limit access to employment opportunities (Avalos et al. 2021; Nakamura and Avner 2018). This issue is examined in various contexts, including public transit connectivity (Iimi 2019; Nell et al. 2025) and traffic congestion.

For example, estimates from Tanzania suggest that worsening congestion could potentially halve job accessibility (Ochoa et al. 2021). Although urban areas are typically seen as drivers of economic growth (Li and Rama 2023), including green growth (Bluedorn et al. 2023), enhancing labor mobility within cities is essential to fully realize their potential.

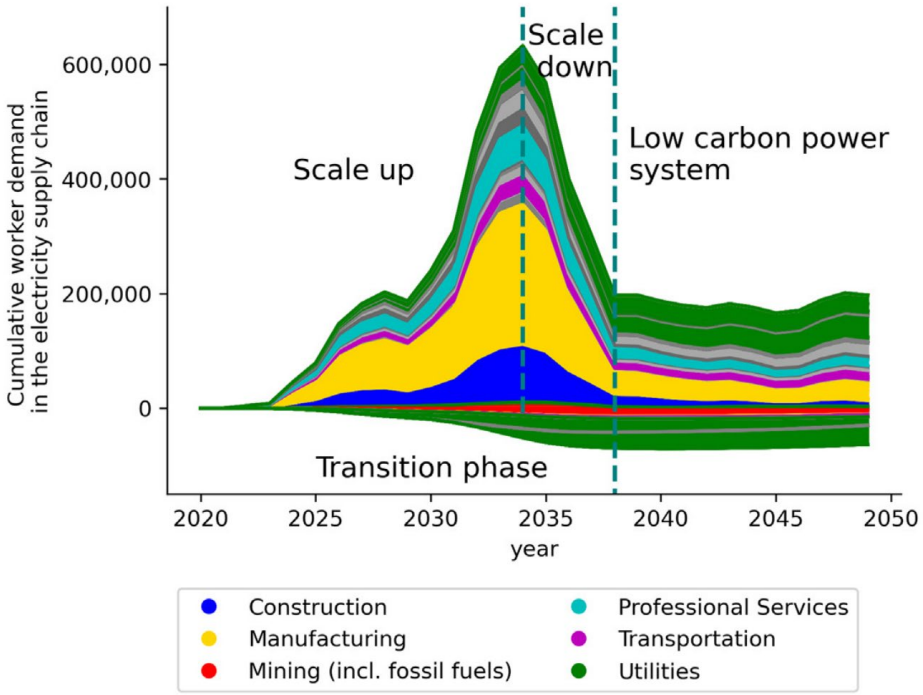
3.3. Temporal mismatches

Temporal mismatches refer to the discrepancies between the timing of labor supply and demand in the labor market. These frictions are particularly pronounced during periods of major economic or technological transformations, where smooth reallocation of workers is crucial. However, studies have shown that the reallocation of workers in response to structural shocks tends to be slow (Donovan and Schoellman 2023). For example, Porzio et al. (2022) find that 40% of the historical decline in agricultural employment across 69 countries is attributable to generational shifts—i.e., new generations entering the labor market in non-agricultural jobs—rather than farmers transitioning out of agriculture. In the context of the green transition, the demand for workers skilled in green technologies is expected to increase, but the existing workforce may lack the necessary training or experience, leading to delays in filling these roles and slowing the overall transition to a greener economy. This mismatch between the timing of green job creation and the readiness of the workforce to fill them highlights the impact of temporal labor frictions on the green transition.

The growing recognition of the green transition as a catalyst for job creation often overlooks the fact that some roles may be needed only for a short duration. For example, Bücker et al. (2025) show that, in a simulated rapid transition scenario of the United States power sector, the aggregate demand for jobs does not follow a linear pattern but rather exhibits three distinct phases (figure 8). The first phase is the scale-up phase, characterized by a significant increase in overall labor demand. This phase sees the largest contribution from the manufacturing and construction sectors, which are crucial for producing renewable energy technologies and deploying the necessary infrastructure. The second phase is the scale-down phase, marked by a decrease in overall labor demand as most of the new replacement infrastructure has been built. The final phase is the steady-state phase, where labor demand stabilizes.

By examining the demand trends for different occupations, Bücker et al. (2025) further identify distinct categories of occupations based on their unique demand trajectories. Occupations classified as "consistent growth" experience increased demand during both the scale-up and scale-down phases of the transition. This group includes only a small number of occupations, including solar photovoltaic installers and wind turbine service technicians. In contrast, the "consistent decline" group sees a decrease in demand during both phases, with occupations in this category primarily employed in mining, extraction, and fossil-fuel operations. The "temporary growth" occupations experience an increase in demand during the scale-up phase, followed by a decline during the scale-down phase. This group encompasses production, construction, and engineering occupations, as well as some installation and maintenance, management, business, and administrative roles. Early identification of potential labor shortages and labor stranding is crucial to enable policymakers to more effectively support workers and employers in addressing these challenges.

Figure 8: An analysis of labor demand trends under a rapid decarbonization scenario in the United States power sector highlights critical temporal dynamics



Source: Bücker et al. 2025.

Another example from the Tajikistan CCDR evaluates the impact of the newly mandated compulsory use of solar power systems in the construction and reconstruction of buildings on workforce requirements (World Bank Group 2024a). The findings reveal that approximately 65,000 workers will be needed in 2025, representing about 2.5% of the 2023 workforce, to meet the new mandate. However, the CCDR also highlights that this surge in worker demand is expected to be temporary. The initial spike is attributed to low labor productivity and high labor intensity as the regulation is rolled out and workers undergo a training and learning phase. Drawing from experience from the United States, the labor intensity of small-scale solar installations decreases rapidly (from about 36 to 18 workers per megawatt between 2018 and 2022). This rapid decline in labor intensity poses a challenge, as the initial high demand for workers will quickly taper off, potentially leading to an oversupply of trained workers and subsequent unemployment in the sector.

Temporal mismatches driven by short-term factors like investment cycles or seasonal shifts can also influence the course of the green transition. For example, the construction of renewable energy infrastructure, such as solar panels and wind turbines, is highly sensitive to the timing of investment flows. As a result, job creation in this sector can fluctuate based on investment cycles and policy continuity (Hanna et al., 2024). In China, for example, the withdrawal of solar subsidies in 2021 introduced investment uncertainty, leading to reduced hiring in clean energy firms (Asim et al., forthcoming). Although these issues are not unique to green technologies—fossil fuel-dependent

technologies are also highly influenced by commodity price fluctuations, for example, where rapid price increases can lead to surges in exploration and production activities, and sharp declines when prices fall—these cyclical mismatches add another layer of complexity to labor market dynamics in the green transition, requiring careful planning and adjustment by both workers and employers.

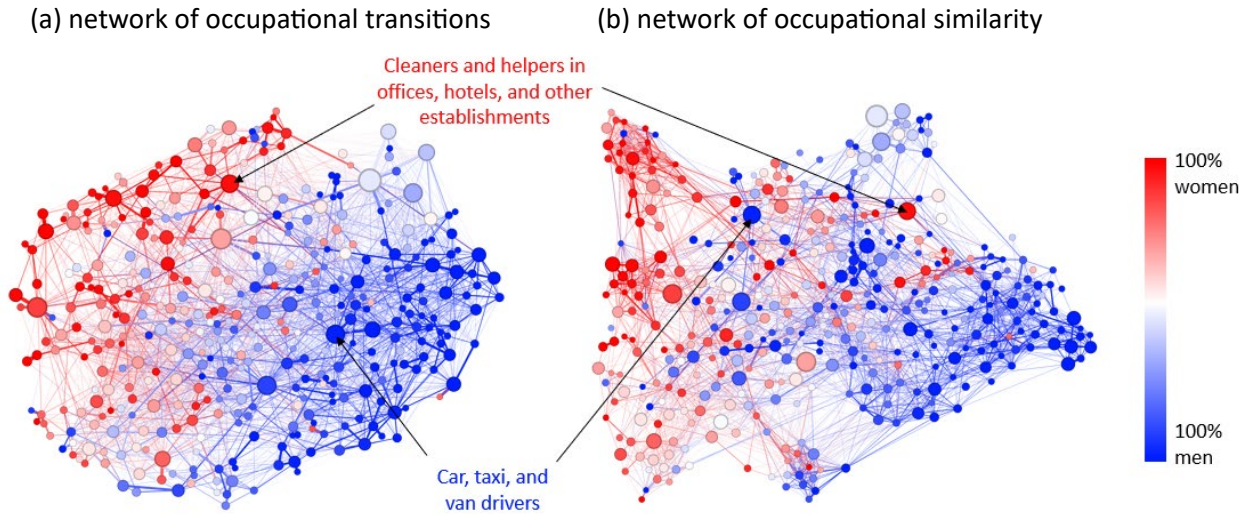
3.4. Norm mismatches

Various factors of social exclusion, including gender and age, can hinder the smooth reallocation of workers. Norm mismatches occur when employment opportunities and job mobility are limited due to discrimination against certain worker characteristics such as race, ethnicity, socioeconomic status, education level, disability status, and other demographic factors. Additionally, job transitions can be affected when workers themselves discriminate against certain types of jobs or industries based on social or cultural biases. Addressing these norm-based mismatches is crucial for economic growth (Chiplunkar and Kleineberg 2025; Hsieh et al. 2019). For example, achieving gender parity in employment and pay could potentially unlock up to a 20 percent increase in GDP per capita globally (World Bank 2024a).

Historical job switching patterns exhibit highly gendered and stereotypical biases, as illustrated in figure 9a. In this network, each node represents an occupation, and the nodes are linked based on the probability of transitioning from one occupation to another using data from Georgia for 2020-2023. Nodes that are located closer together indicate a higher probability of transition, while nodes that are further apart indicate a lower probability. The nodes are color-coded by gender, with occupations overrepresented by men colored blue and those overrepresented by women colored red. In 2023, 54% of occupations in Georgia were at least 85% male or female. The network demonstrates that workers in occupations predominantly held by women tend to transition to other occupations with a similar gender composition, and the same pattern is observed for men. Specifically, between 2020 and 2023, only 3% of workers moving into male-dominated occupations (defined as those with over 85% male concentration) are women, while 7% of those transitioning into female-dominated occupations are men.

Figure 9b presents an occupational network based on task similarity. Similar to figure 9a, figure 9b visualizes the gender distribution across occupations in Georgia. Notably, while both figure 9a and figure 9b are clustered by gender, figure 9b shows a weaker clustering effect. This suggests that occupations requiring similar tasks do not necessarily have similar gender compositions. Therefore, it appears that even when the nature of the work itself is not inherently gender-specific, the transition patterns between occupations can be influenced by social or cultural factors. In other words, figure 9b illustrates that even when occupations require very similar tasks, gender restricts occupational mobility, despite workers having the requisite skills to move from one job to another. For example, sewing machine operators, a predominantly female occupation, have several tasks in common with upholsterers, a male-dominated occupation. These tasks include operating machinery, handling fabrics, ensuring quality control, performing repairs and maintenance, following patterns, and maintaining attention to detail. However, due to social or cultural barriers, workers rarely transition between these two types of occupations.

Figure 9: Gender differences in occupational mobility in Georgia



Source: Developed by the authors.
Notes: Each node represents an occupation and is color-coded to indicate the proportion of female (red) and male (blue) workers. In panel a, the links between nodes reflect the probability of transitioning from one occupation to another. Nodes that are closer together indicate a higher probability of job transitions, while nodes that are further apart indicate a lower probability of transitions. In panel b, the links between nodes reflect occupational task similarity, with more similar occupations positioned closer together and less similar ones further apart.

Many CCDRs also find that emerging occupations in the green transition tend to be male-dominated, potentially limiting female workers' ability to benefit from the increased demand in these fields. For example, the Namibia CCDR reveals that 15 percent of male workers are employed in green hydrogen-related occupations, compared to only 4 percent of female workers (World Bank Group, forthcoming-a).[4] Promoting the green transition without addressing women's lower participation in these transformative sectors risks exacerbating existing gender disparities in the workforce. Furthermore, women tend to exhibit lower mobility compared to men, making them more vulnerable to job displacement during economic transformations (Cardoso and Hartmann 2023). In agriculture, where women in lower-income countries often perform the more labor-intensive tasks, new technologies that reduce emissions and lower labor requirements pose significant transitional challenges for female workers. For example, a study in India indicates that shifting rice production from conventional methods to direct-seeding or machine-transplanting techniques could disproportionately affect women compared to men (Gartaula et al. 2020).

Female workers can also be indirectly affected by the job displacement of men. For example, in the United Kingdom (UK), the closure of coal mines led to a decrease in female employment in nearby manufacturing plants, while male employment in those plants increased (Aragón et al. 2018). This

[4] Due to the nascent nature of the green hydrogen industry, many of its specialized occupations are not yet included in standard classifications. Therefore, for this analysis, we rely on related proxy occupations. For example, although electrolysis engineers are needed, they are not classified separately; instead, the broader category of chemical engineers is included, as they share relevant skills and knowledge.

suggests that men who lost their mining jobs took up positions in manufacturing, potentially displacing women from those roles. However, the extent of these substitution effects can vary depending on the industrial and occupational structure of the area. For example, in Pennsylvania's hard coal region, the share of women in the local labor force increased as men were displaced from mining jobs (Diluiso et al. 2021). This indicates that the impact of male job displacement on female employment can differ based on the specific economic context.

Age is another factor that significantly limits occupational mobility (Curtis et al. 2024). Generally, the returns on investing in new skills and learning new tasks are lower for older workers, while the costs may be higher. As a result, mobility decreases with age, and older workers tend to move between occupations with higher skill similarity (Cardoso and Hartmann 2023). In the Tajikistan CCDR, older workers are found to be more prevalent in occupations essential for the green transition (World Bank Group 2024a). Conversely, the Georgia CCDR indicates that younger workers are more prevalent in digital occupations crucial for the green digital economy (World Bank Group, forthcoming-b).

Ethnicity and race are also significant determinants of occupational mobility. In the United States, certain demographic groups such as Black and Native Americans are significantly underrepresented in jobs supporting the green transition and are less likely to possess the required skills (Kuersteiner and Ordal 2023). This outcome aligns with other fundamental differences in mobility patterns for these groups: for example, Black workers have significantly lower returns to early-career job mobility (Guo 2021) despite having the same tendency to voluntarily leave their first stable job in search of better opportunities (Park and Sandefur 2003). They also face higher earnings losses when switching jobs at a later stage (Guo 2021) and are much more likely to face involuntary job separation than White workers (Park and Sandefur 2003).

People with disabilities face greater challenges to mobility than those without (de Raaf et al. 2023; World Bank 2022b). Additionally, besides facing generally lower employment rates (World Bank 2022c), persons with disabilities are overrepresented in occupations that require lower educational attainment and involve part-time work (Tur-Sinai 2019), and they face significant barriers to accessing training programs (García Mora et al. 2021). As a result, disabled people are less likely to participate in green sectors, which typically demand higher skill levels and offer more formal employment opportunities (World Bank 2023a). These disparities leave workers from already underprivileged groups more vulnerable to job and income losses associated with economic transitions.

3.5. Preference mismatches

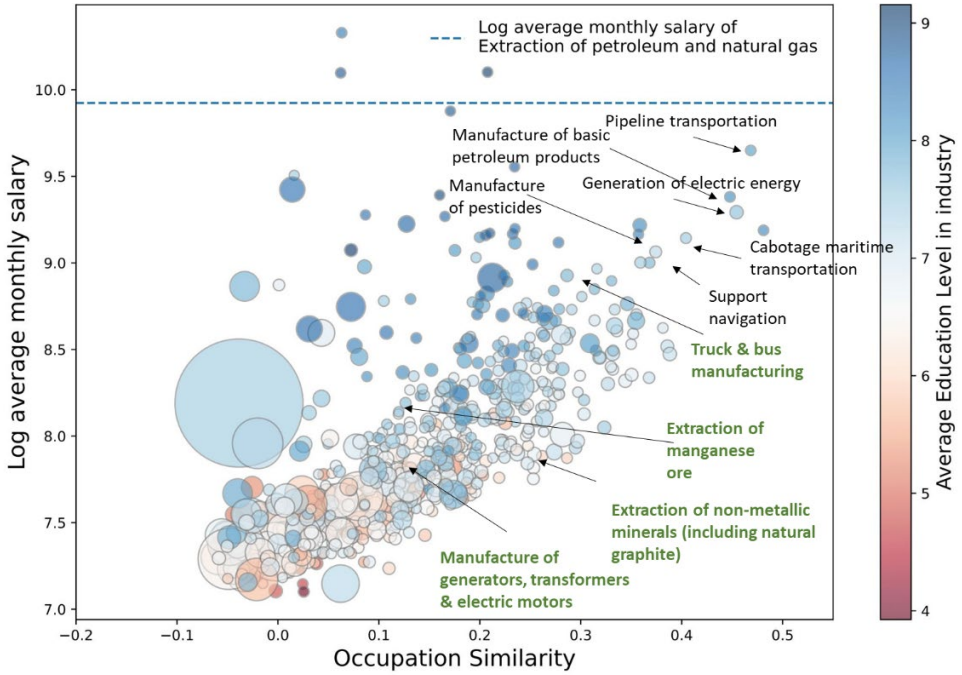
People engage in economic activities for various reasons, most notably to generate income, but also for social and psychological fulfillment. Frictions related to compensation can occur when newly created jobs, such as those supporting the green transition, offer lower wages than existing jobs, thereby disincentivizing workers from transitioning to the new roles. For example, in the Brazil CCDR, workers in the petroleum and natural gas industry were found to have several occupation transition opportunities that involve a high degree of similar tasks (figure 10). However, many of these jobs offer significantly lower wages, making them less attractive to these workers (World Bank Group 2023b). Similarly, the

Malaysia CCDR finds that workers shifting from at-risk oil and gas roles to in-demand electrical and electronics jobs with comparable skill requirements would incur wage reductions of 20%–70%, creating a strong disincentive to transition (World Bank Group, forthcoming-c).

In theory, the labor market would see wages for workers in declining sectors decrease in response to reduced demand, thereby incentivizing a shift to alternative employment sectors. However, in practice, wages do not necessarily adjust, due to factors like long-term contracts and union negotiations. Additionally, behavioral biases, such as anchoring—where workers' wage expectations are tied to their previous earnings—further complicate this transition. Consequently, despite economic signals suggesting a need for labor reallocation, psychological barriers can maintain the status quo.

Salary mismatches can lead to significant short- and long-term income losses for displaced workers. Following a job loss, workers often face persistent and substantial earning losses (Rud et al. 2024). These losses are particularly pronounced if workers lose bargaining power and are forced to accept jobs that are very different from their previous roles and/or offer lower wages to avoid unemployment. To mitigate human capital losses and avoid a decrease in wages, workers switching jobs tend to move into similar occupations (Cardoso and Hartmann 2023). However, this strategy may not be feasible during a large-scale economic transition.

Figure 10: Transition possibilities for workers in the petroleum and natural gas industry



Source: Developed by the authors.

Notes: Each bubble represents a potential target industry for workers transitioning from the petroleum and natural gas industry. The size of the bubbles corresponds to formal employment in Brazil in 2018, and the color indicates the average education level of the industry's formal workforce. The x-axis displays the occupation similarity of each target industry to the petroleum and natural gas sector, while the y-axis shows the average monthly salary for each industry. Occupations labelled in green relate to jobs with greater demand in a green transition. The dotted blue line represents the current average monthly salary for workers in the petroleum and natural gas industry.

Indeed, salary mismatches have had significant impacts in previous coal phase-outs. In the UK, workers displaced from coal mines faced sizable and long-lasting earning losses. Even those who found jobs after being displaced earned, on average, 40% less during the first year after losing their jobs compared to similar workers who were not displaced. Fifteen years later, this group still earned about 20% less on average (Rud et al. 2024). Similarly, as the coal phase-out progressed in Germany's Ruhr area, the alternative industries in low value-added assembly work were not able to offer wages as high as those in the coal industry (Galgóczi 2014). These examples underscore the fact that formal coal mining jobs typically provide higher wages, in part to compensate for the hazardous and demanding nature of the work (Ruppert Bulmer et al. 2021). In contrast, salary mismatches tend to be less significant during the phase-out of the informal coal sector, where wage levels are generally lower and employment conditions less regulated.

Even if workers are offered higher wages to move into green jobs,[5] a preference mismatch could still occur if their primary motivation lies in other factors such as job security and long-term career stability. For example, if a worker is offered a well-paying position in a renewable energy company but is more motivated by the stability and other non-monetary benefits provided by their current job, the financial incentive may not be enough to persuade them to make the switch. A survey in Germany and the Netherlands found that women preferred scheduling flexibility and a company with a good reputation, whereas men preferred jobs with high earnings and a permanent contract (Valet et al. 2021). In Costa Rica, many fishers remain in their traditional work despite available alternatives, due to strong occupational inertia, including community attachment, and other non-monetary factors (Galvis and Rodriguez-Novoa 2024).

4. Overcoming labor market frictions

This section outlines a range of potential policy solutions aimed at addressing the five frictions identified in sections 2 and 3 (table 2). Its purpose is to support policymakers in exploring a variety of options and to encourage further analysis and debate. The proposed solutions span several domains—including education, social protection, industrial policy, and pension reform—and approach the frictions through multiple policy levers: some target supply-side challenges, while others focus on demand-side issues. Additionally, some measures seek to strengthen monitoring or improve the matching efficiency between labor supply and demand. No single intervention is likely to be sufficient; rather, effective responses typically require coordinated policy packages that combine actions across different domains and tackle the frictions from several angles.

Importantly, these frictions rarely exist in isolation. The interactions between the identified frictions are often complex, as each friction can influence and exacerbate the others, underscoring the need for

[5] Evidence on green wage premiums is mixed and context-specific. In the United States and Europe, green-intensive occupations earned a modest premium of about 7 percent over 2005–2019 (IMF 2022). In the United Kingdom, the premium appears larger for lower-skill occupations (Whittard et al. 2025). By contrast, Switzerland shows evidence of a "sustainability wage gap", with lower wages in more sustainable sectors—potentially reflecting workers' pro-environmental preferences (Krueger et al. 2023).

tailored policy packages that consider the multifaceted nature of the challenges faced. For example, as illustrated in Box 1, addressing skill mismatches through training initiatives may not result in successful transitions if frictions related to commute time are not addressed. This example highlights the importance of a comprehensive approach, including worker profiling and individualized support tailored to the needs of the different labor market segments likely to be affected (Honorati and Banaszczyk 2023).

Box 1: Multifaceted obstacles to the successful retraining of coal workers

As ZE PAK, a lignite conglomerate in Poland, began collective redundancies in 2020, they organized specialized training sessions to help dismissed workers transition to roles in the solar industry. Of nearly 300 workers offered the training, 50 registered, and all graduated. After the training was completed, a meeting was organized with a solar energy company, Esoleo, inviting all training participants. At the meeting, Esoleo presented the characteristics of their job offers and invited interested candidates for individual interviews. Only 14 ended up accepting job offers: 5 as photovoltaic systems electricians, 6 as photovoltaic installers, and 3 as construction engineers. Despite the high quality of the training, few workers transitioned to new roles. The main reasons included the long daily commute (up to 120 kilometer) and the advanced age of the trainees, which made it challenging for them to work as installers on a solar farm. This case highlights the need for solutions that go beyond skill development, addressing other barriers such as spatial and preference-related factors.

Source: Honorati and Banaszczyk 2023.

Because each of the five frictions can be driven by different factors—such as market failures or transaction costs—the most effective interventions will depend on accurately diagnosing these root causes within specific contexts. For example, in the case of skills mismatches, if the primary distortion stems from information asymmetry—where workers lack awareness of how their existing skills might transfer to other roles—then policy measures like enhanced labor market information systems or career guidance services would be appropriate. On the other hand, if the main barrier is high transaction costs associated with acquiring new skills, such as expensive training programs or lack of access to reskilling opportunities, then solutions like subsidized training or financial support for upskilling would be more effective. A careful analysis of the specific factors at play is thus essential to designing targeted and impactful policy responses. Given the complexity of the frictions and their drivers as well as the broad scope of the relevant policy areas, the options outlined here are intentionally high-level and not exhaustive. They are intended to serve as a starting point for discussion, rather than as definitive or prescriptive recommendations. Table 2 lists potential solutions, while sections 4.1 to 4.5 provide additional resources for readers seeking a deeper understanding of the available policy approaches and interventions. As illustrated in the second column of table 2, while several of the interventions address market failures, others are motivated by considerations of fairness and distributional equity.

Table 2: Examples of potential policy solutions

Friction	Relevant market failures or rationale for intervention	Policies and interventions	Country examples
SKILL (section 4.1)	**Information asymmetry:** Workers and employers often lack timely, accurate information about qualifications and the skills needed for rapidly changing jobs, leading workers to invest in misaligned skills and employers to struggle finding candidates with the right competencies.	• Establish approaches for identifying, monitoring, and anticipating skill needs • Develop methods for identifying existing skills and their likely evolution • Enhance job matching and career guidance services	A job matching tool in Poland uses machine learning to analyze labor demand data and occupational task descriptions, providing jobseekers with a list of potential jobs available in the local labor market that match their previous work experiences (Honorati et al. 2023).
	Coordination failures: Education and training systems may not coordinate effectively with industries to adjust curricula or programs as job requirements evolve, particularly in the face of rapidly changing needs.	• Strengthen foundational learning and tailor tertiary education and training programs to current and future labor market needs • Attract skilled workers through global strategies	In Germany, the dual education system integrates classroom learning with company-based practical training in close collaboration with industry, contributing to 87% of apprentices receiving permanent job offers from their training companies (Shabangu and Madondo 2024).
	Externalities: Weak institutional capacity in many low-income countries limits formal employment opportunities, reducing demand for skilled workers. This underemployment imposes social costs not borne by firms, potentially resulting in market demand for skilled labor below the socially optimal level.	• Stimulate labor demand through targeted economic policies and incentives	In Romania, a personal income tax exemption for IT workers reduced hiring costs, made software development more attractive, reversed emigration, and resulted in eligible firms having 13% more employees than non-eligible firms four years after its introduction (Manelici and Pantea 2021).
SPATIAL (section 4.2)	**Information asymmetry:** Workers and firms lack complete information about job opportunities and candidate availability in other geographic areas.	• Establish approaches for sharing real-time labor market data on vacancies and skills across geographic areas	In the Republic of Korea, Work-net consolidates data from national surveys, employer postings, and training institutions to provide up-to-date insights on vacancies, skills, and job market trends, allowing jobseekers to search for opportunities locally and nationwide (World Bank 2021).
	Moving costs: Although workers may identify better job opportunities elsewhere, the financial costs of moving and the social costs of leaving established community and support networks often create substantial barriers to relocation.	• Facilitate relocation of workers and firms	In India, the One Nation One Ration Scheme allows cardholders to purchase subsidized food in any locality in the country, making relocation easier without losing access to benefits (World Bank 2025a).

	Regional development and political economy. Migrations often leave the most vulnerable left behind, resulting in social costs and spatial inequalities that are widely regarded as socially and politically unacceptable.	• Align local education and training programs with local demand • Implement place-based development policies	A study in India found that the New Industrial Policy in Uttarakhand led to a 39% increase in the total number of firms and an 82% rise in employment (Li and Roy 2020).
TEMPORAL (section 4.3)	**Dynamic adjustment costs.** Individuals and firms may underinvest in training and reskilling due to uncertainty about future demand and employee retention, leading to a shortfall in flexible skills investment if the social benefits of timely adjustment exceed private returns.	• Implement lifelong learning policies • Promote a more globally mobile workforce	In Korea, the Employment Permit System is a guest worker program aimed at addressing temporary labor shortages, having provided employment to over 540,000 individuals since 2004 (Cho et al. 2018).
NORM (section 4.4)	**Discrimination and behavioral biases.** Labor may not be allocated based on productivity, but instead on prejudice and stereotypes. Since employers and workers do not bear the full social cost of exclusion, this leads to underutilization of talent and efficiency losses.	• Implement and enforce anti-discrimination policies • Invest in education and training programs targeting underrepresented groups • Design employment opportunities tailored to facilitate the participation of marginalized groups • Provide early retirement packages	In South Africa, the Employment Equity Act led firms to expand their recruitment practices, contributing to a US$2,555 rise in women's annual average wages, while men's wages did not change (Landman and O'Clery 2020).
PREFERENCE (section 4.5)	**Loss aversion, fairness, and political economy.** When jobs are lost in sectors like oil and gas, the available alternatives often offer lower wages or fewer benefits. While wage differences usually reflect productivity, they can create political economy challenges, as workers resist lower-paying jobs and see concentrated losses as unfair.	• Introduce wage subsidies and targeted financial incentives • Implement measures to improve job security and stability	In Mexico, an experiment offering wage incentives increased formal employment rates among vocational school graduates by 4.2 percentage points (Abel et al. 2022).

While the policy measures discussed here emphasize targeted interventions to address specific frictions, deeper structural reforms are often necessary to overcome fundamental constraints to growth, such as those related to governance and business regulation. This is particularly critical in countries where policy space is limited and multiple policy objectives need to be balanced. In such contexts, macrostructural reforms can provide policymakers with additional policy levers to facilitate labor mobility and advance the green transition (Budina et al. 2023). Effective solutions thus require comprehensive and well-designed policy packages that reflect the unique, yet often constrained, set of options available in specific country and industry contexts.

4.1. Closing the skill gap

Establish approaches for identifying and monitoring skill needs

The skill gap is a two-sided coin. On one side lies the demand for skills. Workers and education providers often lack timely, accurate, and granular information about the skills needed. Addressing these information asymmetries requires mechanisms for identifying and monitoring the skills required both now and in the future. Various data sources can be leveraged for this purpose, including private sector insights, survey data, administrative data, and real-time labor market information such as job vacancy data. These approaches complement each other; for example, job postings provide near real-time insights, whereas survey data are slower but typically offer more comprehensive coverage of occupations and sectors. Effective coordination among stakeholders—particularly through the incorporation of feedback from employers and industry experts in education and training initiatives—is essential to address coordination failures and ensure that skills identification remains relevant and forward-looking (UNDP IICPSD and SESRIC 2018). A growing number of studies focus on methodologies for identifying skill needs, including those relevant to the green transition. Table 3 lists some relevant studies for readers interested in exploring these options in more detail.

Table 3: Selection of studies on identifying and monitoring skill needs

Source	Overview
Bhutan Labor Market Assessment Report (Alaref et al. 2024)	This report leverages findings from an establishment survey that investigates the recruitment needs of various industries in Bhutan. By analyzing the survey data, the report projects the expected distribution of labor demand by occupation, based on employers' anticipated job vacancies for the coming years.
Skills for a greener future: a global view (ILO 2019)	This report provides qualitative and quantitative analysis of the skills needed for the green transition. It identifies policies and measures to tackle the challenges posed by changes in the demand for skills. With insights drawn from 32 country studies, the report is rich in real-world examples.
Preparing the Workforce for the Green Transition: How to Promote the Green Transition, Competitiveness, and	This report emphasizes the critical role of skill demand assessments, focusing on frameworks in South Africa that provide a standardized language for stakeholders to effectively communicate labor needs. However, the report also addresses the challenges faced in utilizing

Employability by Focusing on Skills Development in South Africa (World Bank 2025b)	these frameworks for skill needs assessments, which are often hindered by a lack of awareness or skepticism regarding their credibility among stakeholders.
Understanding Labour Shortages: The Structural Forces at Play (OECD 2024a)	This report examines how structural factors influence skill requirements and contribute to labor shortages, using data from an employer survey conducted in OECD countries. It finds that firms more affected by the green and digital transitions report higher incidences of shortages. Additionally, companies experiencing shifts in their production technology are more likely to encounter challenges in recruitment.
Understanding Labor Market Demand in Real Time in Argentina and Uruguay (Vezza et al. 2025)	This paper investigates the role of job vacancy data in improving labor market information systems in Argentina and Uruguay, using innovative approaches based on machine learning and artificial intelligence to develop skills classification taxonomies.
The Future of Jobs Report 2025 (World Economic Forum 2025)	This report brings together insights from over 1,000 leading global employers to explore the influence of macrotrends, such as the green transition, on employment and skill demand. It also features analytical work, including on the time required to fill vacancies across different occupations and countries.

Develop methods for identifying existing skills and their likely evolution

On the flip side of the skill gap coin is the supply of skills. Information failures also exist on the supply side, as policymakers and employers often lack timely and detailed data on the current skills available in the workforce and how these may evolve over time. Measuring existing skills and anticipating their future development is crucial for identifying the skill gap. Existing skills can be measured either directly—through standardized testing or skill surveys such as the World Bank's Skills Toward Employment and Productivity (STEP) surveys or the OECD's Programme for the International Assessment of Adult Competencies (PIAAC)—or indirectly through education and training requirements of workers' occupations, which are commonly available in survey data (di Gropello et al. 2011). Data availability for the latter is more ubiquitous, and most of the network analyses presented in section 3 demonstrate this indirect approach.

In addition to identifying existing skills in the workforce, it can be beneficial to evaluate the impact of a changing workforce—such as the retirement of workers with specific skill sets or the entry of new workers—on the skills landscape. Such assessments can assist countries in determining the optimal balance between reskilling current workers and training new entrants. While most countries are likely to implement a combination of these two strategies, the emphasis on one over the other often depends on specific national circumstances. For example, countries with a large youth cohort, like Uganda, may prioritize training new entrants, whereas countries with an aging population, like many European countries or China, might focus more on reskilling existing workers (Caucheteux et al. 2025).

Enhance job matching and career guidance services

Job matching and career guidance services and systems are vital tools for bridging skill gaps by addressing information asymmetries between employers and job seekers (Kadletz et al. 2021; OECD 2021; World Bank 2021; World Bank 2023b). While employment support services can be expensive, recent technological advancements, including digitalization and artificial intelligence, offer promising solutions. For example, a job matching tool developed in Poland uses machine learning to analyze labor demand data and occupational task descriptions, providing jobseekers with a list of potential jobs available in the local labor market that match their previous work experiences (Honorati et al. 2023). Another example is SkillCraft, a skills assessment and career guidance tool available to job seekers in South Africa, which helps individuals map their skills to occupations that best fit their profiles and provides links to job vacancies in the local labor market (World Bank 2022d).

Strengthen foundational learning and tailor tertiary education and training programs to current and future labor market needs

Education and training are critical tools for closing the skill gap, as they equip workers to meet the evolving demands of the labor market. However, market failures such as externalities—where firms underinvest in training because benefits are shared—and coordination challenges among government, employers, and education providers often hinder effective skill development. Addressing these issues requires, first and foremost, investing in foundational education for long-term workforce development and adaptability. It also involves aligning tertiary education programs with changing skill requirements, including those of the green transition, to ensure that new entrants to the labor market possess the necessary skills. Additionally, countries can enhance support for workers in upskilling and retraining to facilitate worker reallocation by expanding adult learning opportunities through short-term education and skills training programs (box 2). There is a substantial body of literature on strategies for bridging the skill gap through education and training. Table 4 lists some of these studies for readers interested in exploring it further.

Box 2: Lessons learned from training initiatives in the European battery sector

The shift to electric vehicles (EV) is expected to significantly impact vehicle manufacturing jobs, with battery production emerging as a primary driver of employment opportunities in the EV sector, a transition that requires new skillsets for workers. The European Union (EU) aspires to become a leader in sustainable battery production and consumption, but the shortage of workers with the necessary skills has become a significant and persistent bottleneck, hindering the rapid expansion of battery production (European Commission 2018). The European Commission has estimated that the battery sector, along with downstream industries such as EV manufacturing and battery recycling, will require the training, upskilling, or reskilling of approximately 800,000 workers by 2025 (European Commission 2023).

Various initiatives have been implemented to address the skills gap in the European battery sector. For example, France and Germany have established battery training schools (McCaffrey and Poitiers 2024) and the European Battery Alliance (EBA) Academy was launched in 2022 (EIT InnoEnergy 2023; 2024).

With a budget of €10 million, the EBA Academy successfully met its target of training 100,000 workers and 100 trainers in less than three years. This initiative provided economies of scale to training for EU countries at a reasonably low cost.

The role of the EBA Academy is comprehensive: it developed a methodology to map the jobs and skills in high demand across the battery value chain, prepared training materials in 10 languages, delivered training at the local level in partnership with Member State institutions, and provided certifications recognized across the EU. This project has now set the path for other 'Net-Zero Industry Academies' in the block, including the European Solar Academy launched in mid-2024 to skill 65,000 workers within two years (EIT 2024).

Challenges to closing the skills gaps in the battery industry still remain, but for the next phases and other industries a few challenges and lessons learned stand out: more investment is needed to close the skills gap, with the greatest demand for shorter vocational courses; the program has not managed to reach older and financially constrained workers, nor people out of employment and education; and translation costs were higher than expected (McCaffrey and Poitiers 2024). On the other hand, the EBA Academy achieved a good gender balance and continues to increase the number of certifications provided.

Table 4: Selection of studies on education and training as strategies for bridging the skill gap

Source	Overview
Quality Early Learning: Nurturing Children's Potential (Bendini and Devercelli 2022)	This report outlines actionable strategies for delivering quality early childhood education at scale in low- and middle-income countries. It emphasizes the transition from evidence-based research to effective policies, aiming to enhance educational outcomes for young children.
Review of the Evidence on Short-Term Education and Skills Training Programs for Out-of-School Youth with a Focus on the Use of Incentives (Clarke et al. 2021)	This paper investigates the effectiveness of training programs targeting out-of-school youth, highlighting the importance of incentives in enhancing outcomes. It underscores the necessity of aligning these programs with labor market needs and employing results-based financing to encourage stakeholders.
Skills at Scale: The Imperative of Tertiary Education for Development (Geven and Bassett, forthcoming)	This report advocates for strategic investments in advanced skills and key reforms in tertiary education systems. It provides an empirical deep dive into the role of tertiary education in delivering advanced skills, drawing on recent household surveys, administrative datasets, and literature reviews. Additionally, it includes guidance on how governments can navigate the political economy of tertiary education reform.
Building Better Formal TVET Systems: Principles and Practice in Low- and Middle-Income Countries (World Bank, UNESCO, ILO 2023)	This report provides an analysis of the challenges and reforms needed in technical and vocational education and training (TVET) systems within developing economies. It emphasizes the importance of aligning TVET systems with labor market demands to enhance employability and productivity. The report draws on global knowledge and experiences, offering practical guidance for policymakers to design and implement effective TVET reforms.

Getting Skills Right: Good Practice in Adapting to Changing Skill Needs (OECD 2017)	This report offers a comparative analysis of the strategies and policies implemented to address skill gaps in five countries: France, Italy, Spain, South Africa, and the United Kingdom. Drawing from the reviews of these countries and the specific recommendations tailored to each, the report presents a collection of best practice principles, many of which focus on education and training, designed to guide the formulation of policies to mitigate skill imbalances.
Choosing Our Future: Education for Climate Action (Sabarwal et al. 2024)	This report provides an overview of how education can equip individuals with the skills needed to facilitate the green transition. It includes a detailed review of existing literature, complemented by new quantitative and qualitative analyses. Additionally, the report outlines a set of short- and medium-term priorities for policymakers, providing strategic guidance to support the green transition.
Human Capital Policy for Development: The Role of Families, Neighborhoods, and Firms (World Bank, forthcoming)	This flagship report explores how human capital, including knowledge and skills, accumulates in three key arenas: families, neighborhoods, and firms. It examines policies that can help individuals in low- and middle-income countries better develop human capital over their lifetimes.

Attract skilled workers through global talent acquisition strategies

Embracing immigration can be a strategic solution to addressing skill shortages (World Bank 2018). A recent study in Croatia, for example, found that the country faces several demographic challenges, such as an aging population and a shrinking workforce, which can be mitigated through well-managed immigration policies (World Bank 2025c). The report highlights how an increasing number of immigrant workers are primarily concentrated in high labor demand sectors such as construction, tourism, hospitality, and manufacturing. However, the report also identifies several challenges associated with immigration and the effective integration of foreign workers, necessitating comprehensive immigration strategies and systems.

Countries can also address skill gaps by entering into bilateral or regional agreements that foster collaboration between nations with complementary labor market needs (World Bank 2023c). For example, global skill partnerships involve origin and destination countries partnering to invest in education and training systems in the origin country to meet skill needs in both countries (Clemens 2015). The destination country—whether through government or private sector initiatives—supports the training of potential migrants to ensure they acquire the qualifications needed by its labor market. This training occurs in the country of origin before migration, benefiting students who remain and enter the domestic labor force (Acosta et al. 2025).

Stimulate labor demand through targeted economic policies and incentives

Although skill mismatches are commonly attributed to an inadequate labor supply, the primary constraint is not always a lack of education or skills (Honorati et al. 2024). Employment in lower-income countries is often concentrated in small and informal businesses, leading to a scarcity of formal-sector jobs. Similarly, in imperfectly competitive markets, wages are often suppressed and job creation is

constrained. In these contexts, job training and job search initiatives alone are insufficient to create substantial new employment opportunities, and policies that can stimulate labor demand—without reinforcing market power in imperfect markets—are essential (Carranza and McKenzie 2024; World Bank Group 2019). For example, the Bhutan CCDR finds that while high-skilled unemployed workers in Bhutan are seeking several green occupations, employers are primarily looking for low-skilled workers, and the vacancies tend to be in non-green occupations (World Bank Group 2025). Achieving the right balance between investing in labor demand and labor supply is crucial for a successful transition. If one advances too quickly ahead of the other, investments will be inefficient. World Bank country diagnostics, such as the Country Climate and Development Reports or the Country Growth and Jobs Report (Country Economic Memorandum 3.0), provide country-specific recommendations and priorities to stimulate labor demand.

4.2. Overcoming spatial challenges

Establish approaches for sharing real-time labor market data on vacancies and skills across geographic areas

The issue of information mismatches—where workers and firms lack complete knowledge about job opportunities and candidate availability—often has a significant spatial dimension. While information may be accessible locally, it can be uneven or asymmetric across different geographic areas. To address this, integrated labor market information platforms are needed to disseminate real-time data on vacancies, skill requirements, and candidate availability across regions. The considerations discussed in the previous section on labor market information systems are relevant here, with the crucial addition that these systems should be designed to inform both workers and firms about opportunities and resources beyond their immediate locality, thereby reducing geographic information gaps.

Facilitate relocation of workers and firms

One strategy for addressing spatial mismatches between labor supply and demand—though often politically contentious and complex—involves incentivizing the relocation of either workers or employers (Grover et al. 2022). Governments can stimulate local economies and create job opportunities by offering tax breaks, grants, or other financial incentives to companies willing to set up operations in regions with high unemployment (World Bank 2011). For example, in Uttarakhand, India, the New Industrial Policy led to a 39% increase in the total number of firms compared with neighboring Uttar Pradesh (Li and Roy 2020). Government incentives for firm relocation are most effective when high relocation costs or market failures—such as limited local information, regulatory barriers, agglomeration externalities, or credit constraints—discourage firms from moving to regions with surplus labor, thereby hindering job creation and economic development.

Underlying drivers of low worker mobility often include the combined financial cost of moving (associated with market failures such as liquidity and credit constraints and housing market rigidities), the social cost of leaving behind established community and support networks, and the loss of non-portable social protection benefits. To address these challenges, financial assistance for moving

expenses, housing subsidies, or tax incentives can encourage relocation to areas with higher labor demand. This approach has been explored in various contexts, such as in Uzbekistan (World Bank Group 2018b). Many countries have also made their social protection systems more flexible, to ensure that migrants keep access to support when they move (Holmes and Branders 2024).

Beyond domestic migration, international migration can serve as an effective tool for addressing spatial mismatches. Research suggests that relaxing restrictions on cross-border movements could yield significant economic gains for both destination and origin countries (World Bank 2018; World Bank 2023c). However, international migration often sparks more controversy than internal migration, due to its complex economic, societal, and human implications.

Align local education and training programs with local demand

Enhancing local education and training programs can help prepare the workforce to meet the demands of the local job market, especially where large-scale government programs have shown limited success (Carranza and McKenzie 2024). A study in Brazil, for example, demonstrated that incorporating feedback on skill demand from local employers can significantly improve the effectiveness and employment outcomes of publicly sponsored job training programs (O'Connell and Mation 2021). Similarly, evidence from Pakistan's renewable energy (RE) sector highlights a geographic mismatch between where RE resources are concentrated and where RE education programs are offered, pointing to the need to expand degree and TVET offerings in high-resource provinces (World Bank 2022a).

Implement place-based development policies

Outmigration can worsen conditions for those who remain. For example, research has shown that outmigration in the United States tends to decrease both labor demand and supply proportionately, which means it does not effectively reduce unemployment for the local residents (Beaudry, Green, and Sand 2018). Place-based policies can be effective to address these issues and spatial mismatches in the labor market, for example when agglomeration effects concentrate productivity and job opportunities in certain regions, leaving others behind. These dynamics often result in persistent regional disparities that also generate difficult political economy challenges.

Place-based policies can include strategies to attract industries or workers, as previously mentioned, but they also encompass a broader range of approaches aimed at addressing barriers to investment in lagging regions (OECD 2025). These approaches typically focus on enhancing the economic infrastructure and attractiveness of underdeveloped or declining regions. For example, investments in transportation infrastructure, such as expanding public transit networks and improving road connectivity, can help reduce the physical distance between where labor is supplied and where it is demanded. In cities, increasing job accessibility is a multifaceted challenge that requires a combination of strategic planning, infrastructure development, policy interventions, and community engagement (Christiaensen and Lozano-Gracia 2023; Li and Rama 2023; Serageldin 2016; World Bank 2023d; World Bank 2024b). Each approach requires careful consideration of local contexts and collaboration between government, private sector, and community stakeholders to ensure sustainable and inclusive development.

Other measures include developing industrial parks, special economic zones, and innovation hubs to attract businesses and stimulate economic activity in targeted regions, although such initiatives can result in both favorable and unfavorable outcomes (UNIDO et al. 2017). However, the costs and benefits of place-based policies depend on their design and scale (Hallegatte et al. 2023). Research indicates that tax incentives alone are insufficient for a policy to succeed (Duranton and Venables 2018). Instead, a combination of interventions such as transportation investments, enterprise development, and innovation support is necessary. These instruments must be well-coordinated to achieve desirable outcomes and should be based on consultations with relevant stakeholders to address political economy challenges (box 3). Although place-based policies have the potential to drive regional development, ex post assessments reveal that these interventions are often costly and yield mixed results (Grover et al. 2022). Rigorous evaluation and ongoing monitoring are therefore essential to ensure efficient use of resources and avoid wasted investment.

Box 3: Lessons learned from phasing out coal in the Ruhr area

The coal phase-out in the Ruhr area, Germany, is renowned for its success in redirecting regional economic activity. When coal production and employment began to decline rapidly in the late 1960s, the state government initiated a development program aimed at economic reorientation. This program sought to attract firms from other sectors, establish universities, and invest in transport infrastructure to enhance worker mobility to new job sites. However, challenges arose as mining companies refused to sell land to new firms, effectively obstructing the government's efforts to diversify economic activity for over a decade, a situation known as the “ground lock”.

Eventually, a bottom-up, multi-level, participative, and city-focused approach resolved the “ground lock”. By 1980, a second program to neo-industrialize the region was implemented. Learning from the failure of the previous top-down approach that led to the “ground lock”, the “Action Program Ruhr” for economic reorientation was built around a dialogue among national, state, and municipal institutions. In parallel, additional mechanisms were employed to acquire land and end the “ground lock”. From the mid-1980s, further development plans for individual cities were enacted. Workers either transitioned to new jobs or exited the workforce through early retirement packages (Herpich et al. 2018).

Nonetheless, rejuvenating local economies in coal-dependent towns remains a difficult task. History has shown that even with substantial financial support, transforming these areas often faces significant challenges (World Bank Group 2018a; Lobao et al. 2021). For example, in the UK coal transition, Rud et al. (2024) found sizable and long-lasting earning losses for all displaced workers. In low-income countries, in particular, helping workers move out of towns that rely solely on coal mining might be a more effective solution.

4.3. Strategic temporal coordination

Implement lifelong learning policies

During periods of significant economic or technological transformation, certain skills may become obsolete. Market failures—such as underinvestment in retraining, either on the part of firms that fear workers will leave after being reskilled or on the part of individuals who are uncertain whether demand for new skills will persist—can prevent workers from adapting effectively. In such situations, the implementation of lifelong learning policies to support worker retraining for new roles becomes essential (ILO 2023a). These policies enable workers to continuously update their skills to meet the evolving demands of the labor market (see section 4.1).

Specifically, during the scale up phase of economic transformation (see section 3.3), lifelong learning initiatives can help equip the existing workforce with newly required skills, thereby minimizing delays in filling critical roles and supporting the smooth implementation of new projects. Conversely, as the transformation progresses and the demand for certain roles diminishes, lifelong learning policies can provide displaced workers with opportunities to retrain and transition into other sectors. This continuous cycle of skill development and adaptation not only addresses immediate labor shortages but also mitigates the long-term impact of job displacement. By fostering a culture of ongoing education and skill enhancement, lifelong learning policies can play a crucial role in managing the dynamic needs of the labor market during periods of major economic change.

Promote a more globally mobile workforce

A theoretically appealing solution to address some of the key temporal labor market frictions of the green transition would be to promote a globally mobile workforce. Facilitating the mobility of workers across borders could help mitigate the impact of fluctuating labor demand (World Bank 2023c). For example, as the demand for construction workers decreases in one region after the completion of a major renewable energy project, these workers could move to other regions or countries where new projects are starting. However, implementing this idea in practice presents numerous challenges. Differences in immigration policies, language barriers, cultural differences, and the complexity of international labor agreements can hinder seamless labor mobility. Furthermore, the recognition of skills and qualifications across different countries can be a significant hurdle, as standards and certifications vary widely.

Despite the barriers to global labor mobility, several regional schemes offer opportunities for workers to move across borders and fill temporary employment gaps. In the Pacific Islands, for instance, temporary labor mobility programs in Australia and New Zealand have become essential sources of income and employment for many islanders (Doan et al. 2023). Similarly, the Employment Permit System in the Republic of Korea is another example of a guest worker program that addresses temporary labor shortages, having provided employment to more than 540,000 individuals since its introduction in 2004 (Cho et al. 2018).

4.4. Inclusive workforce strategies

Norms often reflect long-standing cultural values and societal expectations, which can be resistant to change. Addressing norm-based frictions in the labor market can therefore be particularly challenging. To effectively tackle these issues, comprehensive and integrated strategies are essential, including behavioral interventions and legal reforms.

Table 5 summarizes relevant literature offering policy recommendations to strengthen inclusion. While some of these address underlying drivers—such as social norms—most focus on promoting inclusion without directly confronting these root causes, reflecting an acknowledgment of the enduring rigidity of such norms.

Implement and enforce anti-discrimination regulations and policies

Implementing and enforcing anti-discrimination regulations and policies can help ensure that hiring practices are equitable and transparent, thereby providing equal opportunities for all candidates irrespective of their demographic characteristics (WHO and World Bank 2011). A notable example is the Employment Equity Act in South Africa, which prompted firms to broaden their recruitment efforts across a wider range of sectors and resulted in an increase in the average wage for women (Landman and O'Clery 2020).

Invest in education and training programs targeting underrepresented groups

Providing education and training programs that target underrepresented groups can help individuals from marginalized backgrounds acquire the skills and qualifications needed to compete in the labor market (Pankratova et al. 2024), including in the green transition (Deininger et al. 2023). For example, the importance of skills-based programs in promoting women's economic inclusion is highlighted in a randomized control trial conducted in rural Uganda (Lang and Seither 2024). The study demonstrates that women who participated in a course on entrepreneurship skills are 17% more likely to generate income from their businesses.

Design employment opportunities tailored to facilitate the participation of marginalized groups

Countries can also design employment opportunities that are specifically tailored to facilitate the participation of marginalized groups, such as women in contexts with restrictive gender norms. In Djibouti, the high take-up rate of a public works program designed for women suggest that when suitable opportunities are offered, women are willing to participate in the labor market (Devoto et al. 2024). However, the reversion to non-participation after the program ended indicates that temporary interventions may not be sufficient to sustain long-term employment. Therefore, more permanent and comprehensive measures can be essential, such as providing accessible and affordable childcare services, offering skills training and education programs to equip women with market-relevant skills, enacting and enforcing legal and policy reforms to protect women's rights in the workplace, and establishing social protection programs to support women during job transitions (Ajayi and Koussoubé

2024). A study across eleven emerging market economies demonstrates that integrating gender-focused measures—particularly investments in social infrastructure such as childcare—alongside green transition initiatives not only leads to more balanced employment gains, including increased participation by women, but also boosts overall economic growth. Specifically, these measures can raise GDP by up to 0.3 percentage points compared to scenarios without such interventions (Deininger et al., forthcoming).

Provide early retirement packages

Early retirement can sometimes serve as a viable solution, particularly in contexts where there is an oversupply of labor or where older workers face challenges in adapting to rapidly changing job requirements. These programs can be designed to provide financial incentives and support to older workers, and can be particularly effective in sectors experiencing declining demand. In such cases, early retirement schemes can help manage the transition by gradually reducing the workforce in these sectors while avoiding sudden job losses. For example, in China, to mitigate the effects of labor redundancies affecting tens of millions of workers in the 1990s, the government introduced an early retirement strategy for workers that were within five years of the retirement age (World Bank Group 2018a). However, such strategies can also impose significant financial strain on the pension system and, if not carefully implemented, may result in the loss of valuable human capital for the economy. This is especially the case if used excessively by companies undertaking massive layoffs without productive considerations for the labor market (Marguerie et al., forthcoming).

Table 5: Literature on policies promoting inclusion in the labor market

Source	Overview
Gender-Based Employment Segregation: Understanding Causes and Policy Interventions (Carranza et al. 2019)	This paper explores the persistent issue of gender-based employment segregation, highlighting both horizontal and vertical segregation in various sectors and occupations. It discusses supply-side and demand-side constraints, as well as macro-fiscal, legal, and regulatory frameworks, offering policy levers to address these challenges and promote gender equality in the labor market.
Technical Note on Accessibility (World Bank 2022b)	This note identifies barriers to accessibility and offers strategies to address these challenges, focusing on enhancing access to buildings, transportation, communication, and information for persons with disabilities.
Financing mechanisms for promoting social inclusion in skills and lifelong learning systems: Global overview of current practices and policy options (ILO 2023b)	This report provides a review of global practices and policy options aimed at enhancing social inclusion through skills development and lifelong learning systems. It highlights various financing mechanisms such as performance-based contracts, public-private partnerships, targeted grants, and sustainability-linked loans.
A Case Study of Results-Based Financing for Vocational Education and	This study on results-based financing for vocational education and training highlights the use of performance-based contracts and financial incentives to promote social inclusion.

Training (Zanola et al. 2021)	
Can Job Training Decrease Women's Self-Defeating Biases? Experimental Evidence from Nigeria (Croke et al. 2017)	This paper demonstrates that ICT training significantly increased women's employment in the ICT sector, particularly among those with biases against women's professionalism. This suggests that targeted job training can help overcome internalized biases and facilitate women's entry into male-dominated sectors, offering a promising policy tool for enhancing gender inclusion.
Equality of Opportunity for Sexual and Gender Minorities (World Bank 2024c)	This report recommends the development of inclusive legal and regulatory frameworks to enhance access to labor markets for sexual and gender minorities. It emphasizes the need for countries to adopt laws that protect against discrimination and violence, ensuring that sexual and gender minorities can participate fully and equally in the workforce.
All in My Head? The Play of Exclusion and Discrimination in the Labor Market (Das 2015)	This report highlights the need for comprehensive anti-discrimination policies that address both direct and indirect forms of exclusion in the labor market. It recommends implementing legal frameworks that protect marginalized groups from unconscious biases and systemic discrimination, thereby promoting a more inclusive workforce.
Pathways to Climate-Resilient Economic Inclusion: A Framework for Integrating Climate Action in Economic Inclusion Programs (Costella et al. 2023)	This report outlines how economic inclusion programs can advance labor market inclusion in a green transition through skills and coaching, access to finance, and market linkages—prioritizing women and other climate-vulnerable groups.

4.5. Navigating preference barriers

Introduce wage subsidies and targeted financial incentives

Political economy considerations are central to understanding the challenges faced by displaced workers from sectors such as oil and gas during the green transition (Hallegatte et al. 2023). These workers often encounter alternative employment opportunities that offer lower wages or fewer benefits, making them reluctant to transition into new roles.

Concentrated economic losses among affected groups can fuel resistance to structural change, thereby undermining support for the green transition. When the social benefits of reallocating labor to industries with lower negative externalities are significant, governments can address these political economy challenges by, for example, offering wage subsidies or targeted financial incentives, such as signing bonuses, to employers in green sectors. Such measures can also help address underinvestment in nascent green industries, which may suffer from underinvestment due to positive spillovers such as innovation. From a strict economic efficiency perspective, while such interventions can distort relative wages and may reduce incentives for mobility into higher-productivity sectors, when equity, distributional fairness, or political stability are central concerns, temporary wage support that can help bridge the compensation gap between new and traditional industries and accelerate the development of green sectors can be justified. Importantly, these measures should be temporary and well-targeted, designed to address short-term mismatches and support adjustment, rather than to indefinitely maintain

wages above market rates. Such approaches were particularly effective during the COVID-19 pandemic (World Bank 2023b) and can also be beneficial during the phase-out of fossil fuels (Cunningham and Schmillen 2021). However, evidence also suggests that while wage subsidies can help avert immediate job destruction, they do not necessarily result in long-term retention of employees. For example, during the global financial crisis, wage subsidies in Mexico helped employment rebound faster in eligible industries compared to ineligible ones, but did not necessarily lead to firms retaining workers (World Bank Group 2021).

Temporary income support for displaced workers and incentives for on-the-job search and training can also potentially reduce salary mismatches (Econie and Dougherty 2019; OECD 2024b; Sharpe and Martinez-Fernandez 2021; Sofroniou and Anderson 2021). Unemployment insurance, for example, could increase the probability of upwards mobility and reduce unfavorable occupational switches into lower wage positions as it allows workers to afford searching for a new job for a longer period (Cardoso and Hartmann 2023). However, unemployment insurance could lead to unfavorable outcomes if it hampers reallocation and skill formation in expanding sectors (Porto 2012). Additionally, fully funded external training programs for redundant workers, which can be attended during working hours, coupled with flexible release policies, could aid workers in exploring new job opportunities. This approach ensures that workers do not have to forfeit redundancy packages that are intended to facilitate job transitions, a strategy adopted in the United Kingdom to redeploy coal workers effectively (Stewart 2024).

Implement measures to improve job security and stability

Workers and employers often face significant information gaps regarding the long-term prospects, wage levels, and career stability associated with emerging green sectors. Workers may be uncertain about the viability and sustainability of new roles, while employers may not fully recognize the breadth of transferable skills present in the existing workforce. These information gaps can heighten concerns about job security and career stability, making transitions into green emerging sectors less attractive. To address these frictions, comprehensive career development programs are essential for providing long-term stability in new sectors. Such programs could include training and upskilling opportunities that are aligned with the needs of the green economy, ensuring that workers have the necessary skills to succeed in their new roles (see section 4.1). Additionally, policies that promote job security, such as offering permanent contracts and strong labor protections, can make new jobs more attractive to workers who prioritize stability over immediate financial gains, although such measures reduce employer flexibility and may result in reduced labor demand.

5. Discussion and future work

This paper has introduced a practical framework for diagnosing labor market frictions that impede worker mobility in the green transition. By classifying misalignments between labor demand and supply along five dimensions—*what* workers do (skills), *where* they are (spatial), *when* they are available (temporal), *who* they are (norms), and *why* they work (preferences)—the framework can help policymakers pinpoint bottlenecks in the labor market that can slow the pace of the green transition and

raise unemployment among displaced workers. Drawing on World Bank analyses, particularly Country Climate and Development Reports, it has provided examples of analytical work that illustrate each of the five frictions.

The paper has further highlighted that an enabling mix of active labor market programs, social protection, and education and training reforms is central to converting job creation potential into realized jobs. Managing these frictions is also a political economy imperative: concentrated losses from displacement and spatial divergence can undermine support for green policies unless accompanied by credible, worker-centered transition pathways. Countries that act proactively on these bottlenecks will, all else equal, move faster, face lower adjustment costs, and capture larger and more widely shared benefits. While the framework has been developed in the context of the green transition, it is generalizable to labor market transitions more broadly.

The analytical examples in the third section aimed to quantify and visualize the complexity of each of the five frictions. These examples demonstrate tools that can pinpoint occupations, workers, and regions at greater risk, and others that can identify the most feasible transition pathways. The goal of these tools is to help direct attention and resources to the most pressing bottlenecks. However, existing tools typically cannot quantify how long or costly these frictions will be—for individual workers and for the economy as a whole. Two further quantification steps are therefore important avenues for future research.

First, more data and additional tools are needed to understand the time and resource implications of the five frictions for individual workers and employers. For example, if a worker seeks to transition to an occupation with 80% skill overlap, how long will it take to master the remaining 20%? How many months of formal study would that require, or how much productivity loss during on-the-job training? How much longer does it take to fill a vacancy in a remote location? Answers will depend on job-specific tasks and country context. Data on historical occupational transitions can help but are often insufficient. Berryman et al. (2025), for example, use historical data to model detailed labor flows but do not quantify the specific time and resource costs of different frictions.

Second, incorporating the five frictions into macroeconomic modeling can, for example, help clarify their impact on productive capacity, economic development, the macroeconomic effect of unemployment, and the speed of economywide transition. This will require further information on the “production function” of critical occupations or skills to model industrial output when shortages lead to unfilled vacancies in certain places. It also requires a better understanding of local welfare systems and how long-term unemployment affects the economy through a smaller working population and higher social security spending.

The central takeaway is this: identifying and managing skill-, spatial-, temporal-, norm-, and preference-related frictions is critical to unlocking job growth, maximizing the macroeconomic gains of the green transition, and sustaining public support for reform. While more work is needed, this paper advances the discussion by presenting a consolidated framework and practical analytical tools to inform policies and programs that address these frictions and support workers through the green transition.

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